Pritesh Rodge

+91 9028029773 priteshrodge.work@gmail.com LinkedIn Github Website

Education

Indian Institute of Information Technology, Design and Manufacturing Jabalpur, M.P
Bachelors of Technology in Computer Science and Engineering *Aug. 2023 - present*

Experience

Robotics Software Intern - UGO Inc. Jun 2026 - July 2026
Summer Intern 2026 *Tokyo, Japan - Hybrid*

- Engineered edge-based digital meter reading pipelines using **YOLOv8n** and **PaddleOCR**, leveraging hardware **NPU & optimizations** to achieve a **43x speedup** and reduce total inference time to **under 50ms** for reading 7-Segment digital meters.
- Overhauled a modular analog meter reading architecture by superseding heavy models to their mobile counter-parts and **migrating model to NPU**, resulting in a **97.38% reduction in execution time from over 304 seconds to under 8 seconds** for analog gauge meters.
- Engineered a cross-framework model conversion pipeline, translating **PyTorch** and **PaddleOCR** models to **ONNX** and subsequently to **.adla** (Amlogic Deep Learning Accelerator) format via a proprietary SDK, enabling seamless deployment and execution on **NPU** hardware.
- Researched and applied model optimization techniques including **quantization**, **pruning**, and **knowledge distillation** to reduce model size and latency while preserving accuracy for edge deployment.
- Resolved severe 4K camera detection failures at a 1-meter distance by implementing systematic detect-segment-recognition approach, achieving **over 98% final reading accuracy** across both digital and analog systems.
- Developed an interactive Quick-Train application with a guided labeling interface, enabling on-site users to seamlessly capture 30 to 40 image datasets and **auto-train models** to recognize **unseen meter types**.

Linux Kernel Mentorship (LFX) - The Linux Foundation June 2025 - August 2025
Linux Kernel Bug Fixing Summer Mentorship 2025 *Remote*

- Debugged **Linux kernel crashes** by reproducing Syzbot-reported issues using **Syzkaller, QEMU, and GDB**.
- Traced **memory corruption and overflow bugs** through low-level code analysis and **runtime debugging**.
- Developed and submitted **kernel patches** via mailing lists, incorporating reviewer feedback to meet **upstream standards**.
- Gained deep understanding of **kernel internals - system calls, scheduling, and memory management** through hands-on debugging.

Projects

Senseye - Navigational Aid for Visually Impaired Individuals (Patent Pending)
Jetson Nano, Luxonis RGB-D, ESP32, LRA Haptics, BLE/Wi-Fi Drive

- Built a **fully functional wearable assistive product** for visually impaired users, comprising a **waist-mounted depth sensing unit** and a **footwear sleeve with directional haptic feedback** for real-world use : hands-free, audio-free, and network-free navigation.
- Engineered an **end-to-end perception pipeline** on Jetson Nano using a stereo RGB-D camera, performing **real-time 3D obstacle detection** and lateral zone classification (left, center, right) entirely on-device, with **distance-proportional vibration intensity** delivered through I2C-multiplexed Linear Resonant Actuators on the shoe unit.
- Designed and fabricated the complete **dual-module hardware system** -including a **custom actuator driver circuit** with **voltage-level adaptation** and galvanic safety isolation between the ESP32 controller and LRA haptics, a thermoplastic rubber footwear sleeve, and a Wi-Fi wireless link between modules - from schematic to working prototype.

Autonomous Warehouse Drone - eYRC 2024-25
ROS 2, Gazebo Fortress, Python, C++, PID Control, Path Planning, Image Processing

- Built an **autonomous nano-quadcopter system** to navigate warehouse aisles, detect packages, and stabilize precisely at target setpoints in a **dynamic environment**.
- Implemented a custom PID controller achieving stabilization within an error margin of ± 0.4, enabling accurate **waypoint navigation** and drone control Video
- Used **ROS 2** nodes with **Gazebo simulations** for localization via **WhyCon markers**, Mapping, and efficient **Path-planning** Video

Structure from Motion (SfM)
A Structure-from-Motion Implementation from Scratch GitHub

- Built a **Structure-from-Motion pipeline from scratch** to reconstruct 3D scenes from 2D image sequences, implementing core stages without relying on high-level SfM libraries.
- Worked on understanding **first principles of computer vision** - camera geometry, feature matching, and multi-view reconstruction to build a deeper understanding of each stage of the pipeline.

Technical Skills

- **Languages:** C/C++, Python, SQL, Java.
- **Libraries & Frameworks:** PyTorch, ONNX, OpenCV, YOLOv8, PaddleOCR, Pandas, NumPy, scikit-learn, Matplotlib.
- **Systems & Tools:** Linux, Git, Docker, QEMU, GDB, ROS 2, Gazebo, NPU Deployment (Amlogic ADLA).
- **Coursework / Concepts:** Computer Vision, Machine Learning, Data Structures, Operating Systems, Computer Networks, DBMS, OOP.

Skills / Leadership / Volunteering

- Ranked Top **25** among **700+** teams nationally in the **e-Yantra Robotics Challenge (eYRC 2024-25)** through development of an autonomous warehouse drone system.
- Led robotics system development for national competitions including **Robocon 2025**, coordinating design, testing, and integration across sub-teams.
- Mentored **300+** school students through hands-on robotics workshops, simplifying complex technical concepts into practical demonstrations.